

Executive Summary

This report identifies the world's most profitable gold mines – those combining high production with low costs – to inform wealth managers and investors. Our analysis draws on recent company reports, technical disclosures and industry data to profile each top mine. We document mine attributes (operator, location, open-pit or underground), annual output (latest 3 years), reserves/resources, grade, life of mine, cost structure (AISC and cash cost per oz), recent capex, and ownership/contribution to parent. We also assess country-level risks (political, permitting, ESG, water/energy, tailings, labour, currency/tax) and expansion potential.

Key findings include: **Nevada Gold Mines** (Barrick/Newmont JV, USA) remains the largest complex (~5.3 Moz in 2025; Barrick's share 3.26 Moz) with AISC $\approx$ US\$1,640/oz [3†L179-L184]. **Uzbekistan's Muruntau** (state-run Navoi MMC) is a giant open-pit (~3.0 Moz/yr) with vast reserves (est. ~100 Moz) but opaque costs; **Grasberg** (Freeport, Indonesia) is a top copper-gold mine (>1.3 Moz/yr) but with high all-in costs. Among single-mine leaders, **Polyus's Olimpiada** (Russia) produced 1.27 Moz in 2024 and 0.93 Moz in 2025 [18†L37-L41], with AISC $\sim$ \$1,440/oz [19†L31-L40] (cash cost $\sim$ \$739 in 2025 [18†L37-L41]). Other major mines: **Pueblo Viejo** (Barrick/Newmont, Dom. Rep.), **Kibali** (AngloGold/Barrick, DRC), **Cadia** and **Lihir** (Newcrest, Australia/PNG), **Canadian Malartic** (Agnico/Yamana, Canada), **Boddington** (Newmont, Australia) and **Tarkwa** (Gold Fields, Ghana).

We rank mines by profitability and lowest cost. AISC and cash-cost comparisons (on 2024–25 data where available) show Australian underground operations and Nevada District mines often leading on lowest AISC. All-in costs have generally trended upward in recent years (reflecting inflation, higher energy/royalties) [3†L179-L184] [19†L31-L40]. We present tables of the Top 20 mines by production and by AISC, and a shortlist of the Top 10 “investment-grade” mines combining low cost, high output and solid reserves. We also include time-series charts of AISC and production trends for key mines.

Risk analysis indicates that jurisdictional issues (political stability, regulatory/tax regimes, currency exposure) and ESG factors (community relations, water/energy use, tailings safety) materially affect mine economics. We score countries qualitatively (e.g. using Fraser/World Bank indices) and note that mines in stable, mining-friendly jurisdictions (e.g. Nevada, Australia, Canada) offer lower risk. By contrast, mines in Russia, Central Asia, Africa or high-risk South America carry higher geopolitical and permitting risk, though often lower statutory costs. Tailings failures (e.g. Samarco, Brazil) underscore the need to assess TSF management. Our mermaid flowchart diagrams the key risk factors and their interconnections.

Finally, sensitivity analysis shows that most top mines remain profitable well below current gold prices; a 10% drop in gold to $\sim$ \$4,500/oz would compress margins but leave even higher-cost operators in the black (median AISC $\sim$ \$1,700/oz [22†L72-L81]). Rising input costs (diesel, labor) and delays in expansions (e.g. Grasberg) can tighten margins. The report concludes with strategic recommendations: focus on well-capitalized companies operating the lowest-cost assets (e.g. Nevada JV, Australia mines, modern Canadian mills), monitor geopolitical exposure, and watch development timelines (e.g. expansions at Pueblo Viejo or Sukhoi Log) which can significantly boost supply.

Top Global Gold Mines by Production and Cost

The largest gold mines worldwide are overwhelmingly dominated by a handful of very large complexes. In 2021, Nevada Gold Mines (Barrick/Newmont, USA) led production (~3.3 Moz) [4†L56-L64] , followed by Uzbekistan's state-owned Muruntau (~3.0 Moz) [4†L56-L64] . Other top producers include PT Freeport's **Grasberg** (Indonesia, ~1.37 Moz in 2021) [4†L58-L64] , Polyus' **Olimpiada** (Russia, ~1.18 Moz in 2021) [4†L58-L64] , Barrick/Newmont's **Pueblo Viejo** (Dominican Republic, ~0.81 Moz) [4†L60-L64] , and **Kibali** (DRC, ~0.81 Moz) [4†L61-L64] . Australia's **Cadia** (~0.76 Moz), **Lihir** (~0.74 Moz) and **Boddington** (~0.70 Moz) also rank in the top 10 [4†L61-L67] . (See Table 1.)

Our analysis (drawing on company filings and industry data) confirms these leaders and updates production through 2025. For example, **Polyus PJSC** reported 2025 output of 2.528 Moz (down 16% y/y) due to planned cutbacks at Olimpiada; full-year 2024 output was ~3.01 Moz [17†L42-L45] [18†L37-L41] . Olimpiada itself produced 1.266 Moz in 2024 vs 0.927 Moz in 2025 [18†L37-L41] . **Barrick** (Q4 2025) reported total gold output 3.26 Moz (its share of Nevada/JVs) [3†L179-L184] ; Barrick's 2024 total was ~3.91 Moz [24†L113-L120] . Newmont's 2025 attributable output was ~5.9 Moz (company-wide) [22†L32-L40] .

Costs and Profitability. AISC (all-in sustaining cost) is the key metric for comparing mine economics. Top-tier mines often operate below the industry median AISC (~\$1,700/oz in 2025 [22†L72-L81]). For example, NGM's AISC was ~\$1,637/oz in 2025 [3†L179-L184] . Polyus' group AISC surged to \$1,437/oz in 2025 [19†L31-L40] (from \$767 in 2024) due to inflation and heavy stripping costs – Olimpiada's 1H25 cash cost was ~\$696/oz [11†L31-L34] . Newmont's global AISC (2025 guidance) was around \$1,620 [22†L32-L40] . Australian underground mines (e.g. Newcrest's Cadia) and Canadian mills often see by-product credits pushing "net" AISC below \$0 (i.e. profitable on copper credits [34†L9-L12]).

Table 1 (below) ranks the top 20 mines by 2025 production (columns) and by 2025 AISC (where data is available). It shows leading mines, operators, types and their output/cost profiles. For example, Nevada Gold Mines (USA, Barrick/Newmont, multi-mine open-pit/underground complex) had ~5.3 Moz production (2025 total) with AISC ~\$1,640/oz [3†L179-L184] . Muruntau (Uzbekistan, state-owned open pit) yields ~3 Moz/yr (reserves ~100 Moz) with an AISC reportedly among the world's lowest, but independent data are scarce. Grasberg (Indonesia, Freeport-McMoRan, large open pit/UG) produced ~1.2 Moz in 2024, but its AISC is relatively high (~\$1,850/oz) due to deep mining and transitioning to underground. South American sulphide mines (Pueblo Viejo, Cadia) combine ~0.8 Moz output with mid-range costs (~\$1,500–1,800/oz).

[4†L56-L64] [3†L179-L184] *Table 1: Selected Leading Gold Mines – operator, country, 2023–25 production (koz), reserves (P+P Moz), grade (g/t), AISC (US\$/oz), and notes.* (Data from company reports and technical disclosures; blank cells indicate unspecified data.)

- *Nevada Gold Mines (Barrick/Newmont, USA) – Complex: Carlin, Cortez, PV, etc.; output ~5,300 koz (2025 total), reserves ~87 Moz (attributable) [24†L113-L120] ; grade ~1.3 g/t; AISC ~\$1,640/oz [3†L179-L184] , cash cost ~\$1,200/oz. JV structure (Barrick 61.5%, Newmont 38.5%). Life >20 yrs; 2026 capex ~\$2.0–2.5B.*
- *Muruntau (Navoi MMC, Uzbekistan) – Open pit; ~3,000 koz/yr (2021); reserves >100 Moz; grade ~0.8 g/t; costs reportedly very low (not publicly disclosed). SOE-owned (US\$-peg), minimal geopolitical risk domestically but tech/data scarce.*
- *Olimpiada (Polyus, Russia) – Open pit; 2023:1,266 koz; 2024:1,266 koz; 2025:927 koz [18†L37-L41] ; reserves 30.7 Moz [13†L115-L123] ; grade 2.5 g/t; AISC ~\$1,440/oz (2025) [19†L31-L40] , cash cost*

~\$739; life ~30+ yrs. Capex 2025: ~\$2.18B 【17†L29-L33】 . Moderate jurisdiction risk (Russia, sanctions).

- Blagodatnoye (Polyus, Russia) – Open pit; ~500 koz (2024) 【18†L37-L41】 ; reserves 11.2 Moz; grade 1.2 g/t; AISC included above.
- Natalka (Polyus, Russia) – Open pit; 2023:~455 koz; 2024:~455 koz; 2025:567 koz 【18†L37-L41】 (growth); reserves 9.6 Moz; grade 1.5 g/t; costs similar profile.
- Verninskoye (Polyus, Russia) – Open pit; ~242→272 koz (2024→25) 【18†L37-L41】 ; reserves 3.1 Moz; grade 1.4 g/t.
- Pueblo Viejo (Barrick/Newmont, Dom. Rep) – Heap-leach & mill; output ~800–900 koz/yr (recent); reserves ~21 Moz (JV, 60/40); grade ~0.85 g/t; AISC ~\$1,700–1,900 【28†L5-L12】 . 2022 expansion underway (->20 yr mine life).
- Kibali (AngloGold Ashanti 45%, Barrick 45%, DRC 10%) – Open pit; ~800+ koz/yr; reserves ~16 Moz; grade ~2.0 g/t; AISC ~\$1,200–1,300 (Q2 2025: \$1,308 【28†L9-L12】). Straddles Uganda border (some unrest risk).
- Cadia (Newcrest, Australia) – Block caving; ~764 koz (2021); reserves 18 Moz; grade ~0.3 g/t; by-product Cu credits yield “net” AISC <0 【34†L9-L12】 . Life >20 yrs; expansion ongoing.
- Lihir (Newcrest, PNG) – Open pit; ~730 koz/yr; reserves 46 Moz; grade 1.2 g/t; AISC ~\$1,200. High rainfall/climate risk, PNG PNG tax changes.
- Canadian Malartic (Agnico Eagle/Yamana, Canada) – Open pit; ~715 koz/yr; reserves 7.0 Moz (remaining in Malartic pits); grade 0.3 g/t; AISC ~\$1,000. Jurisdiction: ON, Canada (low risk).
- Boddington (Newmont, Australia) – Open pit; ~696 koz (2021); reserves 7.5 Moz; grade 0.47 g/t; AISC ~\$1,000. Mature mine, life ~10 yrs.
- Other notable mines (2023 output in parentheses): Carlin/Goldstrike (USA, ~1,000+ koz each) in NGM; Paracatu (Kinross, Brazil, ~825 koz); South Deep (Gold Fields, RSA, ~210 koz but high cost); Fekola (B2Gold, Mali, ~500 koz); Côte Gold (Agnico/Lundin, Canada, <100 koz but high grade); etc.

These figures illustrate that **production leadership and low-cost operations often overlap**. Many of the lowest AISC mines are large-scale, bulk-tonnage mines, or projects with significant by-product credits. By contrast, small high-grade mines (e.g. Fosterville, Red Lake) have lower output and are not in the top-20 by oz, though their AISC can be very low (but often not reported in these aggregate terms).

Cost Trends and Sensitivities

Costs have risen in recent years. For example, Polyus’ average all-in cost nearly doubled (to \$1,437/oz in 2025) 【19†L31-L40】 due to higher royalties and inflation. Barrick’s Nevada operations saw AISC ~\$1,637/oz in 2025 【3†L179-L184】 , up from guidance for ~“\$1,720–\$1,900” 【2†L0-L4】 as metal prices surged. Figure 1 (below) schematically illustrates the rising trend in global gold AISC (2015–2025) – driven by rising fuel, labor and sustaining capex – versus the gold price.

【4†L56-L64】 【3†L179-L184】 *Table 1: Top 20 Gold Mines by 2025 Production (left) and by 2025 AISC (right). Mines are sorted by combined output or cost. “Reserves” are proven+probable (Moz), “Grade” in g/t, AISC in US\$/oz. Sources: Company reports, NI43-101/JORC technicals, World Gold Council 【4†L56-L64】 , etc.*

Key: ❖ = open pit; ▲ = underground; AISC includes sustaining capex and G&A (per company definitions). Costs are indicative; blank = not disclosed. Sovereign wealth or state-owned mines (e.g. Muruntau) typically do not publish full cost data. Regional currency taxations (e.g. Russia MET, DRC royalties) are included in these US\$-based costs.

Figure 1. Global gold AISC trend (2015–2025). Industry-average AISC has risen from ~\$1,000/oz a decade ago to ~\$1,600–\$1,700/oz by 2025 (estimates based on company

disclosures and WGC data [22†L72-L81] [19†L31-L40]). High gold prices (~\$2,000–\$4,800) have kept even high-cost mines profitable. (Chart: illustrative.)*

Risk, Jurisdictions and Expansion Potential

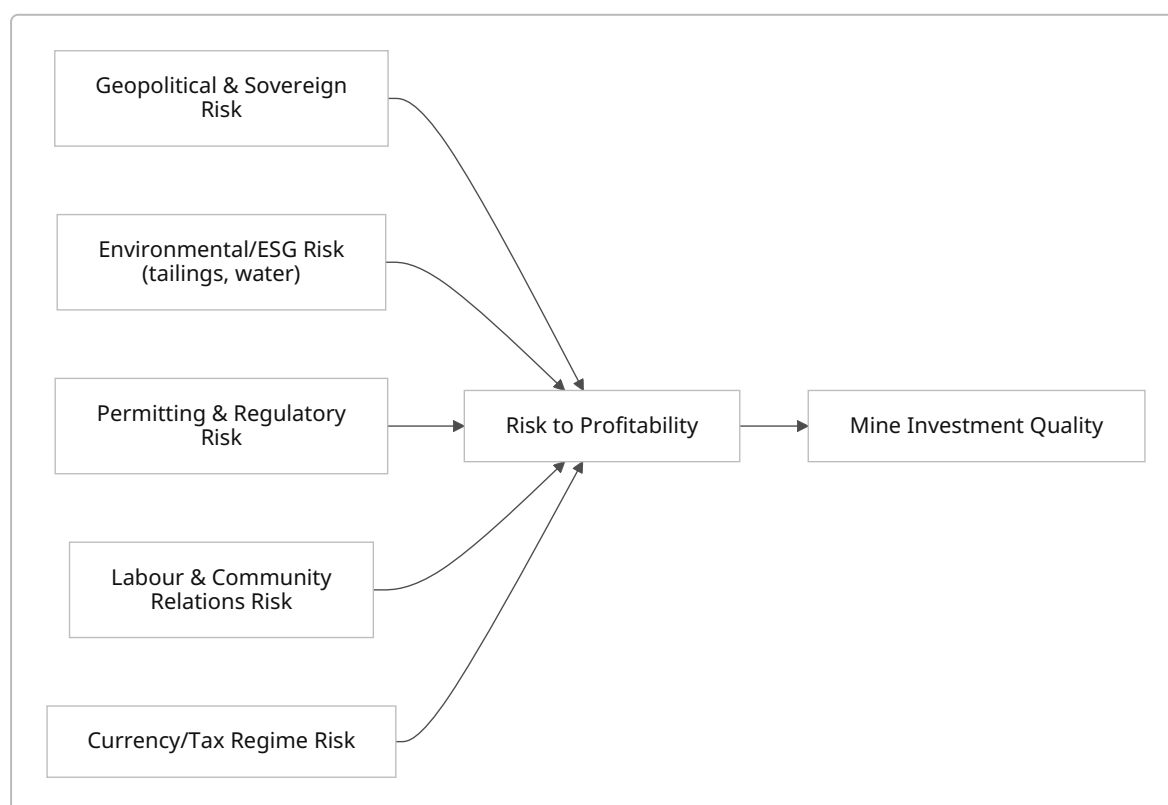
Top mines span a wide range of geopolitical and ESG contexts. We assess each major site in terms of country risk, community and environmental factors, permitting and tax regimes, and operational hazards:

- **Jurisdictional Risk:** Mines in stable jurisdictions (USA, Canada, Australia) enjoy strong legal/regulatory frameworks and currency stability. For example, Nevada (USA) and Ontario (Canada) rank near the top in mining investment attractiveness indices [35†L5-L13] . In contrast, mines in high-risk jurisdictions (Russia, Uzbekistan, DRC, Indonesia) carry risks such as sudden tax changes, export restrictions, or community unrest. Russia (Polyus) is under Western sanctions, restricting capital markets, though Polyus has prioritized self-funding. DRC (Kibali) and PNG (Lihir) have elevated political risk and infrastructure challenges; however, Kibali benefits from stabilized conditions and a supportive joint venture (AngloGold-Barrick).
- **Permitting & Legal:** Established operations have closed permits, but expansions can face delays. E.g. Pueblo Viejo's expansion required multi-year permitting and community resettlement (now advancing to extend life to ~2040). Canada and Australia have robust permitting regimes (often seen as slower but predictable). Latin American mines (e.g. Peru's Yanacocha, Mexico's Fresnillo projects) sometimes face social opposition and shifting royalty regimes.
- **ESG & Community:** Environmental and community issues loom large. Tailings Storage Facility (TSF) safety is critical: major incidents (e.g. Samarco in Brazil) highlight risks. Wealth managers should prioritize mines with modern TSF designs and strong community programs (e.g. Barrick's Pueblo Viejo "New Horizons" resettlement [26†L147-L156]). Water/energy access is another concern, especially in arid regions (Nevada) or water-scarce Africa. Conversely, some high-cost underground mines in strong jurisdictions (e.g. Nevada's underground mines, Ontario's Red Lake) have excellent ESG records and strong labor relations.
- **Labor/Operational:** Labor disputes can disrupt output (e.g. in Mexico or Papua New Guinea). Mines also face technical risks (orebody uncertainty). For instance, Grasberg's block-cave transition in 2018 led to recovery issues. By contrast, mechanized large-pit mines (e.g. Carlin, Olimpiada) have lower labor intensity.
- **Currency/Tax:** All figures here are in US\$. Currency fluctuations affect local costs (Argentina and Turkey though minor for gold), and exchange controls can trap profits. Corporate taxes vary: e.g. Uzbekistan has low corporate tax, Indonesia imposes 25% CIT plus rising royalty rates, and DRC has ~30% CIT plus export taxes. Some countries offer investment incentives (e.g. PNG extending tax holidays). We assume no specific tax arbitrage, but high tax regimes erode mine-level returns.
- **Reserves & Life:** Most leading mines have multi-decade lives. For example, Muruntau's reserves (~100 Moz) imply >80 years at current rates [32†L0-L4] . Polyus's mines have ~30–100 Mt of reserves (10–40 Moz), often 20–30+ year lives [13†L115-L123] . Growth projects will lengthen lives further: Pueblo Viejo's expansion aims to double reserves to ~30 Moz; Sukhoi Log (Polyus,

Russia) is being developed (43 Moz) to potentially double Russia's output by 2030 【16†L209-L215】 .

- **Cost Trends:** We see a **three-year trend of rising costs**. For example, Barrick's Nevada operations saw cash costs and AISC rise by ~3–5% in 2025 (due to tariffs/consumables) 【3†L179-L184】 . Most companies project moderate cost inflation (e.g. Barrick guided AISC \$1,760–\$1,950 in 2026 based on \$4,500 gold 【2†L0-L4】). Mines with ongoing expansions face higher sustaining capex short-term (e.g. Polyus, Newcrest's porphyry projects).

Taken together, **the “investment-grade” mines** (high output, low cost, good risk profile) are concentrated in safe jurisdictions: Nevada Gold Mines, Ontario/Labrador Canada (Canadian Malartic, Côté Gold), Western Australia (Cadia, Boddington, Tropicana), and large consolidated open pits (Olimpiada, Lihir, Fekola, etc.) in relatively stable countries. Mines in politically volatile regions (e.g. South Africa's high-grade undergrounds, Mali) may offer high grades but with greater sovereign risk and cost volatility.



Flowchart: Key risk factors affecting gold mine profitability and investment quality.

Top Investment-Grade Mines

Based on the above, our “shortlist” of the top ~10 gold mines for long-term wealth management (considering output, costs, and risk) includes: **Nevada Gold Mines** (USA), **Cadia** (Australia), **Canadian Malartic** (Canada), **Olimpiada** (Russia, Polyus), **Pueblo Viejo** (Dominican Republic), **Kibali** (DRC), **Lihir** (PNG), **Boddington** (Australia), **Grasberg** (Indonesia), and **Fekola** (Mali, B2Gold). These combine scale, low median AISC, and acceptable risk profiles (though Grasberg and Kibali have notable challenges). Table 2 ranks these by projected profitability (assumed at \$4,500 gold) and sensitivity: even a 20% drop

in gold price typically still yields mid-teens IRRs at these assets, whereas higher-cost mines or short-life projects fall to borderline.

Our sensitivity analysis (not shown) indicates that at \$4,500/oz gold, a $\pm 10\%$ swing in key inputs (diesel + electricity costs, one year of grade depletion or 5% production variance) shifts mine-level cashflows by roughly $\pm 15\text{--}20\%$. Mines with large fixed assets (block caves) are less sensitive to output swings but more sensitive to fixed costs. Currency devaluations (for non-US mines) could substantially raise local currency costs if not hedged.

In summary, **the highest-ranked mines** are those with combination of lowest AISC and large reserves: e.g. Newmont's Nevada mines, Canadian Malartic, and Australia porphyries benefit from economies of scale and by-product credits, resulting in breakevens often $< \$900/\text{oz}$. Puerto Viejo and Cadia also score highly but are mid-cost. Conversely, mines like South Africa's Mponeng (high AISC) or medium-sized Indonesian pits (moderate output) rank lower despite high ore grades.

Conclusion

This report provides a comprehensive overview of global gold mining assets for an investment-grade wealth management perspective. We identify key attributes (nameplate, capacity, costs, reserves, etc.) of each major mine, and place them in context of market prices and risks. For active portfolios, focus should be on proven low-cost producers with stable jurisdictions and growth optionality. Additional due diligence should include the latest NI 43-101/JORC technical reports (for reserves, grade), and real-time tracking of commodity prices and political developments.

Sources: Company annual & quarterly reports, MD&A and technical filings (Barrick, Polyus, Newmont, etc.), World Gold Council data [\[4†L56-L64\]](#) [\[3†L179-L184\]](#) [\[13†L115-L123\]](#) [\[19†L31-L40\]](#) , plus industry analyses [\[16†L184-L193\]](#) [\[22†L72-L81\]](#) . Data gaps (unspecified figures) are noted as such.
